The Boundaries of Evidence-Grounded Trust in Multi-Agent Scientific Debate, with a Controlled Benchmark for Sycophantic Consensus

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Abstract

Large language models can answer scientific questions fluently, but a multi-agent debate over those answers can end in sycophantic collapse: a correct minority agent abandons its answer under pressure from a confident but incorrect majority. Recent systems already check claims against retrieved evidence and reweight agent influence, so the open question is not whether evidence or trust can be used. The open question is when claim-level evidence-derived trust changes the collective outcome, and where its verifier-accuracy and evidence-redundancy limits lie. We describe a deliberation framework that decomposes each agent answer into atomic claims, retrieves scientific passages for each claim, assigns supported, contradicted, or unverifiable verdicts through a verifier interface, and maintains a bounded trust weight per agent. The trust weight is accumulated in the log domain, mapped through one softmax, and projected onto a simplex with a floor and a ceiling; support and harm are normalized by the number of claims checked. The final answer is a trust-weighted plurality, which reduces to majority voting when the weights are equal. The design analysis states four boundary conditions: equal weights recover majority voting; a verification-accuracy threshold $q^\star$ separates correction from amplification; a redundancy boundary $\rho^\star$ limits what more verification can recover; and a shared-error floor bounds the selective risk that any independent verifier can certify. We report the implementation status of the P0 pipeline rather than efficacy: the 13-check gate passes all 13 checks, the current suite passes 523 tests (514 at the P0 release), and Gate 0 completed 10 of 10 vanilla debates with 90 of 90 non-empty positions. A pre-registered evaluation on GPQA-Main and MMLU-Pro STEM, with a 24-run matrix and 1,280 GPU-hours, is specified but not yet run. This version makes no claim that the mechanism reduces sycophantic collapse.

1. Introduction

Large language models are capable of answering scientific questions in a way that could make them useful for tutoring and supporting decisions. However, even though a model may be fluent in its language, it can still provide an answer based on false premises or assumptions. Multi-agent debate can improve factuality over a single pass, but it does not guarantee a correct collective answer. Three agents respond to the same question and debate one another's reasoning over multiple rounds [1]. In addition, chain-of-thought prompting allows users to see how each model arrives at an answer and what steps were taken to arrive at that answer [2]. The weakness lies in the final round of the debate. Most systems use some type of majority vote to determine which answer(s) prevail. Each agent has equal weight, and a response receives credit if it matches the majority [3]. As a result, when a confident wrong answer dominates, the correct minority agent can drop its answer under that pressure, and both sides are then recorded as agreeing on the wrong response [4]. We call this event sycophantic collapse: a correct agent abandons its answer under pressure without decisive new evidence [5]. Yao et al. formalize inter-agent sycophancy in multi-agent debate and show that it collapses disagreement into premature consensus [6]. Sharma et al. show that preference training makes models more likely to accept plausible user positions [7].

Du et al. show that debate improves factuality and reasoning over a single pass, but their agents are treated equally and their claims are never checked against external sources [1]. Pitre et al. report that the correct answer is present in

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the discussion yet ignored in more than 20 percent of wrong-answer cases, and their prompt-rewriting fix still leaves the final decision on self-reported confidence [5]. Yoffe et al. reweight agent messages from token-level uncertainty, and their ground-truth experiment bounds how much a model can improve on that signal [8]. ReConcile pools multiple models and weights each by its internal confidence estimate [9]. Fan et al. target the token cost of debate with a classifier that decides whether a debate is needed at all [10]. Sparse topology reduces communication cost, but it does not reduce the cost of aggregation [11]. Wang et al. combine many models with static aggregation and report strong results on open benchmarks, but the aggregation has no per-agent trust or evidence check [12]. Diversity and training-time methods change how agents reason, not how much each agent's vote counts in the final answer [13, 14]. Intervention testing shows that most sycophancy fixes are not enough on their own to be successful [15]. He et al. find that a previously correct minority is suppressed after a debate; their post-hoc majority reversal occurs after the debate and therefore cannot protect the minority during the rounds [16].

Across these systems, the influence of an agent is set by a vote, a self-reported score, or an internal uncertainty signal. Recent work also checks claims against retrieved evidence, but it does not tie that evidence to a bounded per-agent trust weight inside the debate. The gap is therefore one of measurement and boundary conditions rather than the absence of a mechanism: it is not known when claim-level evidence-derived trust changes the collective outcome, when it amplifies the wrong majority, and how verifier accuracy and evidence redundancy set that boundary. Using self-reported confidence as the influence signal is inadequate because training models to be agreeable also trains them to be sycophantic [17], and alignment experiments show that agreement with wrong user positions increases after fine-tuning [18]. Theory shows that agents converge toward a wrong majority unless an outside correction enters the loop [19], and multi-agent systems fail in recurring ways that diagnosis alone does not fix [20]. Surveys map the design space and its challenges [21]. Applied variants adapt debate to grounded reasoning about knowledge [22] and scientific code generation [23], and other studies examine research ideation [24], educational reasoning [25], and communications engineering [26]. This study ties the influence of each agent to external evidence during the debate. Claims are evaluated against passages retrieved from scientific sources [27], in line with work that grounds correction in external tools [28] and studies that report factual gains from knowledge-grounded correction [29]. The trust weight is bounded by construction, and the final decision uses those weights. Scientific question sets provide a suitable testbed because they carry verifiable answers and published literature that retrieval can access; GPQA and MMLU-Pro provide comparison against established baseline results. This paper reports the design and the P0 implementation status; it does not report efficacy results.

The scope of the study follows from this design. The framework targets scientific question answering, where claims can be checked against a real literature corpus and where objective answers allow final accuracy and correct-minority retention to be measured. Three heterogeneous model families run as served inference behind an OpenAI-compatible interface, and no model is fine-tuned, so the method transfers to any model that can be served under the same interface. GPQA [30] and MMLU-Pro [31] carry objective answers and form the primary comparison; Humanity's Last Exam is an abstention and ceiling control; and a planned gold-evidence set (EG-SSS) will test the mechanism at the claim level. Adversarial sets provide a separate stress test. The study stops at question types where an answer can be verified and an external corpus exists; open-ended and opinion domains, where no claim can be confirmed against evidence, remain outside the scope of the current work. The contribution is a bounded, evidence-grounded trust mechanism together with a reproducible injection and evaluation protocol that the community can reuse to measure sycophantic collapse in other multi-agent systems, whether or not those systems adopt the trust mechanism itself.

1.1. Related Work and Positioning

Three lines of work are closest to this study. The first is in-debate influence reweighting. DebUnc reweights messages from token-level uncertainty [8]. R^2-MAD estimates per-agent reliability from retrieved debate experience and uses confidence weights to modulate peer influence under a shared misconception [32]. To Trust or Not to Trust maintains message-level and agent-level numeric trust under malicious inputs [33]. These systems establish that in-debate reweighting is feasible. They do not ground the weight in claim-level external scientific evidence, and they do not measure the verifier-accuracy boundary at which reweighting stops helping.

The second line is claim-level evidence. GAVEL binds atomic subclaims to explicit evidence units and validates citations through a neutral scrutineer [34]. DebateCV uses a moderator to weigh evidence in debate-driven claim verification [35]. TERA-RAG combines temporal query conditioning, source-reliability-aware evidence scoring, and debate conflict resolution into a reliability-weighted consensus with auditable traces [36]. These systems establish

claim-level evidence binding and reliability-weighted consensus. They do not report when a bounded per-agent trust weight corrects a wrong majority, nor where evidence redundancy defeats it.

The third line is wrong-majority pressure. ConsensAgent detects copying and stalling and rewrites the task prompt after a trigger [5]. Minority Sentinel detects a correct minority after the debate and flips the majority post hoc [16]. Peacemaker or Troublemaker formalizes inter-agent sycophancy in multi-agent debate [6]. These works establish the failure and several mitigations. They do not provide a controlled, pre-registered test that separates the effect of evidence information from the effect of evidence-derived influence, nor a boundary analysis for when that influence helps.

The positioning is therefore a measurement and boundary contribution, not a new mechanism. We do not claim that evidence-grounded debate or trust reweighting is new. We test whether a claim-level, evidence-derived trust weight changes the collective outcome relative to the same debate with the same evidence and majority aggregation, and we pre-register the conditions under which the effect should reverse.

2. Conventional Method

2.1. Baseline Method

The baseline is the multi-agent debate loop with majority-vote aggregation, as introduced by Du et al. [1]. It is selected as the baseline because it is the standard form that the proposed approach modifies, and because several recent systems reviewed in this paper keep majority aggregation while changing the surrounding parts. Chain-of-thought prompting gives each agent a way to expose its intermediate steps [2, 37], and multi-agent debate extends this by letting each agent read the rationale of the others [1]. The baseline treats all agents as equal and combines their final answers with a simple majority vote [4]. Two confidence signals are usually available inside such a loop, self-reported confidence and token-level uncertainty, but neither of them verifies whether a stated claim is true [8].

2.2. Working Principle

The baseline runs in rounds. In the first round, every agent answers the question independently and produces a reasoning trace. In each later round, every agent reads the positions and rationales of the other agents and may revise its own position. After the fixed number of rounds, the loop stops and the final answer is chosen from the last positions. The input is the question, the intermediate process is the sequence of revision rounds, and the output is the answer that the vote selects. The baseline has no source of information that is external to the debate, so the only signals that move an agent are the other agents' arguments and its own confidence.

2.3. Technical Formulation

Let p_i be the position of agent i in the last round and let N be the number of agents. The baseline aggregation rule counts the positions and returns the one with the most votes:

$$a^* = \arg\max_a \sum_{i:\, p_i = a} 1. \tag{1}$$

In this rule each agent contributes one unit of weight regardless of the strength of its evidence, and the vote ignores the individual trust of the agent. This is the exact form that the proposed method replaces with a weighted sum over trust, so Equation (1) is the direct counterpart of the aggregation given later in Equation (5). The baseline has no trust term because it has no evidence signal to build one from.

2.4. Algorithm / Procedure

The baseline follows a fixed procedure. First, each of the N agents answers the question and records a rationale. Second, for each revision round up to the round cap K, each agent reads the other positions and revises. Third, once the rounds finish, the position with the highest head count is selected by Equation (1). The procedure is deterministic apart from sampling, and its cost grows linearly with the number of agents and the number of rounds.

2.5. Characteristics of the Baseline

The baseline is simple, cheap relative to evidence-grounded methods, and easy to reproduce, which is why it remains widely used. It needs no external data source and no additional infrastructure beyond the model server, and it improves over a single pass on many reasoning tasks [1]. These properties make it a fair and stable reference point for the proposed method.

2.6. Limitations and Research Gap

The baseline has one central weakness. Majority voting counts repeated answers and never checks whether a single answer is accurate [4]. If enough agents repeat an incorrect assumption, they can pull one or more additional agents toward it, and the recorded agreement is then called consensus [5]. A confident but incorrect majority can therefore drag a correct minority agent away from its answer, which is the sycophantic collapse that this study targets. Existing work addresses parts of the surrounding problem without touching the aggregation rule: Du et al. [1] treat all agents equally, Fan et al. [10] decide whether to debate but keep the majority vote once the debate starts, Wang et al. [12] use fixed layering with no per-agent trust or evidence check, and Zhou et al. [38] select which agent acts but do not weight its answer. Methods that target sycophancy directly also stop short: Pitre et al. [5] rewrite the prompt but still decide by self-reported confidence, Chen et al. [18] and Ibrahim et al. [17] work at training time on a single model, and He et al. [16] flip the majority only after the debate, which is too late to protect the minority during the rounds. Methods that change the information available to agents face the same limit: Yoffe et al. [8] weight influence from internal uncertainty whose value is bounded by the signal quality, Gou et al. [28] correct a single agent in isolation, Kushwaha and Madhavan [39] score source credibility before one generation pass, and Karpukhin et al. [27] supply the retrieval method used later but no trust weight. Theory and analysis confirm the gap: Estornell and Liu [19] prove that similar responses lead to majority tyranny and that shared misconceptions lower accuracy as more agents adopt them, Choi et al. [4] show that most of the gain credited to debate comes from the vote itself, Cui et al. [40] remove the consensus requirement and score the trajectory instead, Pitre et al. [41] diagnose process quality without changing the decision, Cemri et al. [20] report failure rates from 41% to 86.7% and leave correction to later work, and Cau et al. [42] and Yeow et al. [43] show how persuasion and feedback spread agreement and early errors. Across all of this work, influence is set by a vote, a self-reported score, a rewritten prompt, or a post-hoc flip. The closest systems retrieve evidence and reweight agents, but they do not report a controlled test of when the resulting trust weight changes the collective outcome, nor a boundary analysis for when it fails. That is the gap the proposed method addresses with a bounded trust weight updated from retrieved evidence during the debate and used for the final aggregation.

3. Proposed Method

3.1. Overview of Framework

This is a deliberation layer placed between a question and its answer. A confidence gate either answers the question directly or sends it to a debate with three independent heterogeneous agents. Each agent tags the claims in its reasoning; each tagged claim is checked against retrieved scientific evidence; and the verdicts update the trust weight of the agent. Agents then revise their positions with access to the other positions and their current trust weights. The loop runs for at most three rounds. The final answer is the trust-weighted plurality of the final positions. The result package contains the answer, the citations that support or oppose it, the trust trajectory, and the reasoning trace of each agent. Figure 1 shows the end-to-end flow, and Figure 2 shows the failure mode that motivates the design: a wrong majority can pressure a correct minority, while evidence-grounded trust can reweight the correct minority.

3.2. Confidence Gate

The gate keeps the cost low on easy questions. One agent answers the question with a short chain of thought and reports a confidence value, or three quick agent calls answer the same question and the gate compares the answers. A low confidence or a disagreement sends the question to the full pipeline. A high confidence sends the question to a direct answer. The gate affects throughput and not the core evaluation, because the evaluation set keeps only questions that produce divergent initial answers.

3.3. Heterogeneous Debate Agents

Three agents answer the question in round zero. Each agent is a different model family, so the agents do not share the same training data and the same blind spots [19]. The development configuration uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512. The final configuration replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. All models are served through vLLM behind an OpenAI-compatible interface, and no model is fine-tuned. Each agent produces an answer, a reasoning trace, and a list of tagged claims in the form `<claim id="c1">`. The tags make claim-level checking possible in later stages.

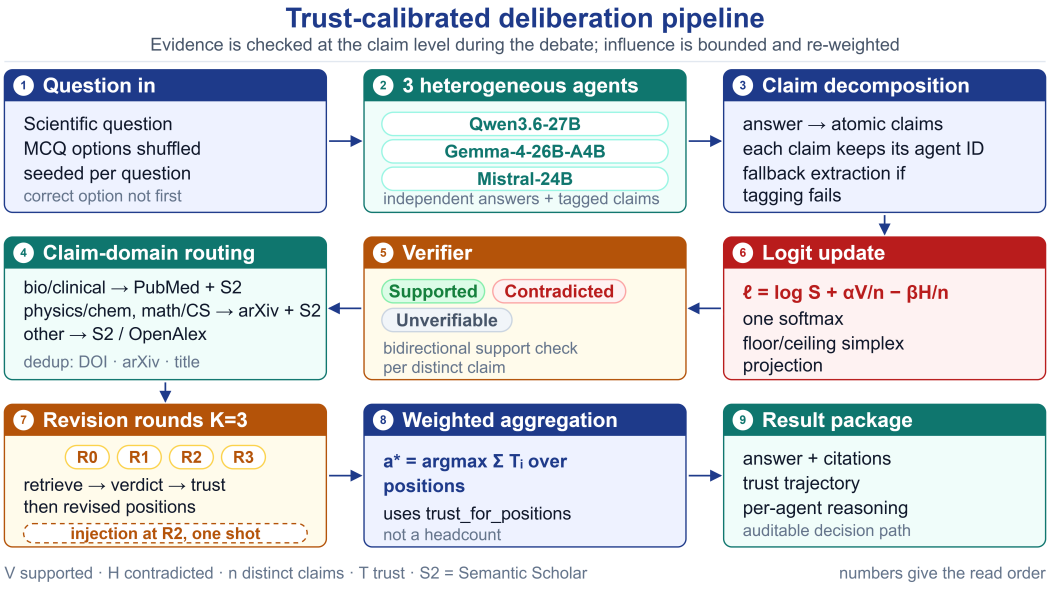


Figure 1: End-to-end flow of the trust-calibrated deliberation framework. The current pipeline renders seeded options in every prompt, routes claims by domain, assigns supported, contradicted, or unverifiable verdicts, updates trust in the log domain, projects it onto a floor/ceiling simplex, and returns the trust-weighted plurality.

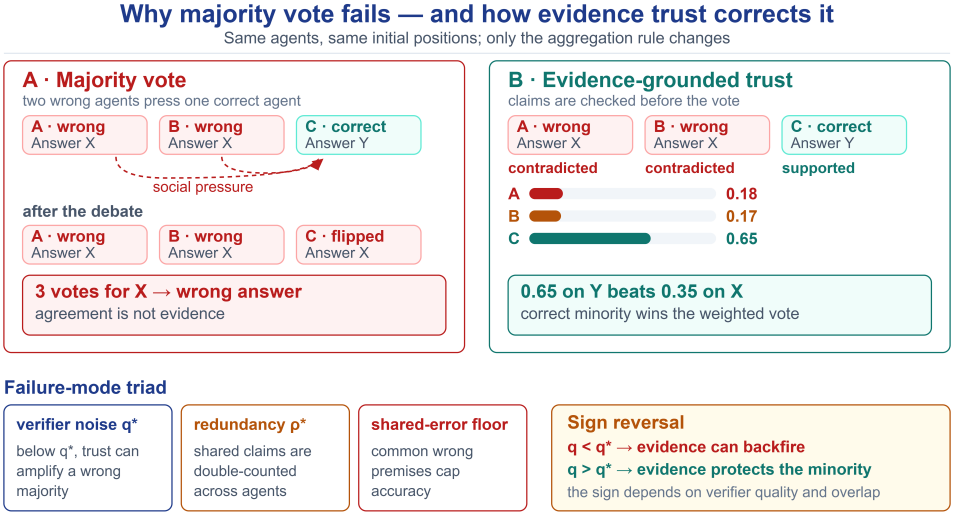


Figure 2: Problem context. A wrong majority can pressure a correct minority into agreement; evidence-grounded trust reweights the correct minority. The failure-mode triad (verifier noise, claim redundancy, shared error) and the resulting sign reversal are shown below.

3.4. Answer Rendering and Scoring

Multiple-choice items are rendered with a seeded per-question permutation, and the permutation is stored in the record. Every prompt that receives a question carries the rendered options; free-text items carry an explicit "Answer format: free text" block instead of fabricated options. The answer parser prefers an explicit answer span, parses caret and LaTeX scientific notation as one number, strips units before numeric comparison, and maps a letter back to the rendered option. A 50-case parser corpus with gold parses provides the regression check, and the parser version is stored with

each record. The options and the parser are prerequisites for every accuracy, collapse, and retention endpoint: without them, a free-form answer cannot be mapped to the option that was shown to the agent.

3.5. Claim Decomposition

A compound sentence can carry more than one fact. The decomposition step splits each agent answer into small factual statements, and every statement keeps the identifier of its source agent. A schema check validates the tagged output. A failed check triggers a fallback extraction pass that recovers the claims from the raw text. A claim that no source can verify is marked unverifiable and contributes neither a supported nor a contradicted count to the trust update.

3.6. Claim-Domain Retrieval and Evidence Scoring

The retriever routes each claim by its domain rather than by the identity of the agent that produced it. Biomedical and clinical claims query PubMed and Semantic Scholar; physics, chemistry, mathematics, and computer-science claims query arXiv and Semantic Scholar; other claims use Semantic Scholar with OpenAlex as a fallback. Routing by claim domain avoids the confound in which one model family is always paired with one source. Retrieved results are de-duplicated by DOI, arXiv identifier, or title, and cached so repeated runs do not hit rate limits. Dense retrieval supplies the passage candidates [27].

A verifier interface maps each claim and its passages to one of three labels: supported, contradicted, or unverifiable. The interface has lexical, cross-encoder, NLI, and LLM-judge implementations. The lexical implementation is the current default; the other implementations are optional and require their own dependencies. The verifier records its name, scores, and evidence identifiers with each verdict. Figure 3 shows one claim checked against a retrieved passage.

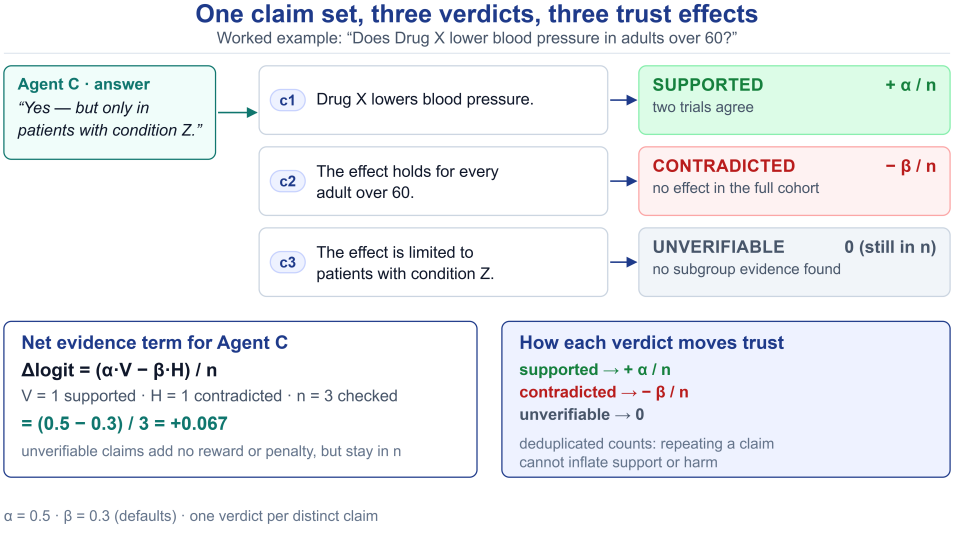


Figure 3: Worked claim decomposition. One scientific claim is split into atomic claims; the verifier labels them supported, contradicted, or unverifiable, and only the decisive labels move the agent's trust.

3.7. Trust Update

The trust update converts the verdicts into a bounded influence weight. Let $s_i(t)$ be the trust of agent i at round t, V_i the number of supported claims, H_i the number of contradicted claims, and n_i the number of claims checked. The update accumulates evidence in the log domain:

$$\ell_i = \log s_i(t) + \alpha \frac{V_i}{n_i} - \beta \frac{H_i}{n_i}, \tag{2}$$

where α and β control the gain and the loss, and the evidence term for agent i is zero when $n_i = 0$. One softmax maps the logits to positive weights:

$$\tilde{s}_i = \frac{\exp(\ell_i)}{\sum_{j=1}^{N} \exp(\ell_j)}. \tag{3}$$

The weights are then projected onto the simplex with a floor f and a ceiling c:

$$T_i = \Pi_{\{w:\, f \le w_i \le c,\ \sum_j w_j = 1\}}(\tilde{s})_i, \tag{4}$$

where Π is the Euclidean projection, computed by water-filling. The projection guarantees $\min_i T_i \ge f$ and $\sum_i T_i = 1$. The ceiling is relaxed to $1/N$ only when $Nc < 1$, where no feasible vector exists. Neutral evidence ($V_i = H_i = 0$ for every agent) is an identity for an already feasible trust vector. Dividing by n_i cancels verbosity: multiplying V_i, H_i, and n_i by the same factor leaves T_i unchanged. The operator is implemented and unit-tested; the debate loop that supplies n_i and recomputes trust from the final positions is reported in Section 5. Figure 4 shows one update.

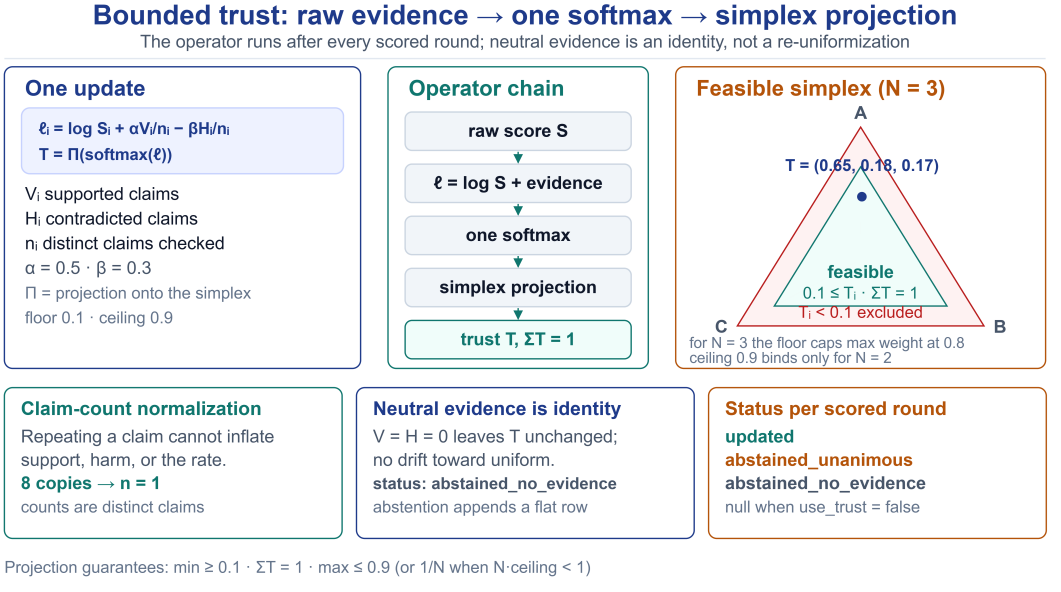


Figure 4: One trust update. Verdict counts are normalized by the number of claims checked, accumulated in the log domain, and mapped through one softmax; the simplex projection then enforces the floor and the ceiling.

3.8. Deliberation Protocol

One debate runs through positions, decomposition, retrieval, verdicts, and trust update. In the three-agent, three-round configuration, a debate makes nine model calls: three initial positions and three revisions in each of two revision rounds. Gate checks add up to three calls when the confidence gate is enabled. Claim checking, trust updates, injection, and aggregation make no model calls in the default lexical configuration. Agents see their own position, the positions of the other agents, and their current trust weights in rounds two and three. The debate stops after round three.

In the stress test, the evaluation harness delivers a fabricated expert consensus once, between round one and round two, and does not repeat it. The injected consensus contradicts the answer held by the correct minority. No consensus is inserted during normal operation. An agent that times out or returns an unparseable response is marked inconclusive for that round, and the debate continues with the remaining agents. Figure 5 shows the round loop, and Figure 6 shows the record and endpoint data flow.

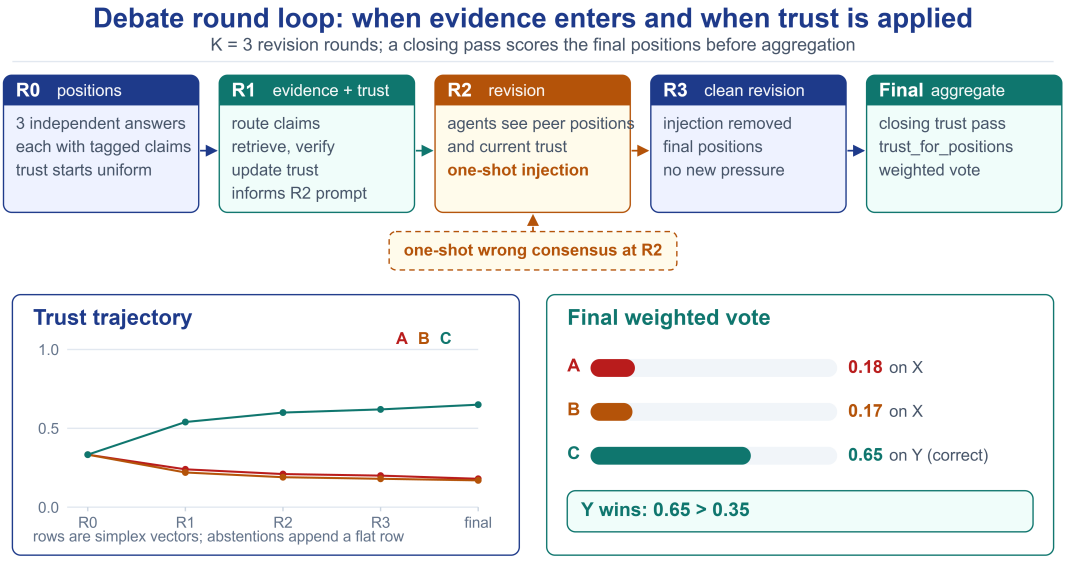


Figure 5: The debate round loop with claim-domain retrieval and one-shot injection. The stages run in order for at most three rounds; the injection enters once between round one and round two in the stress test.

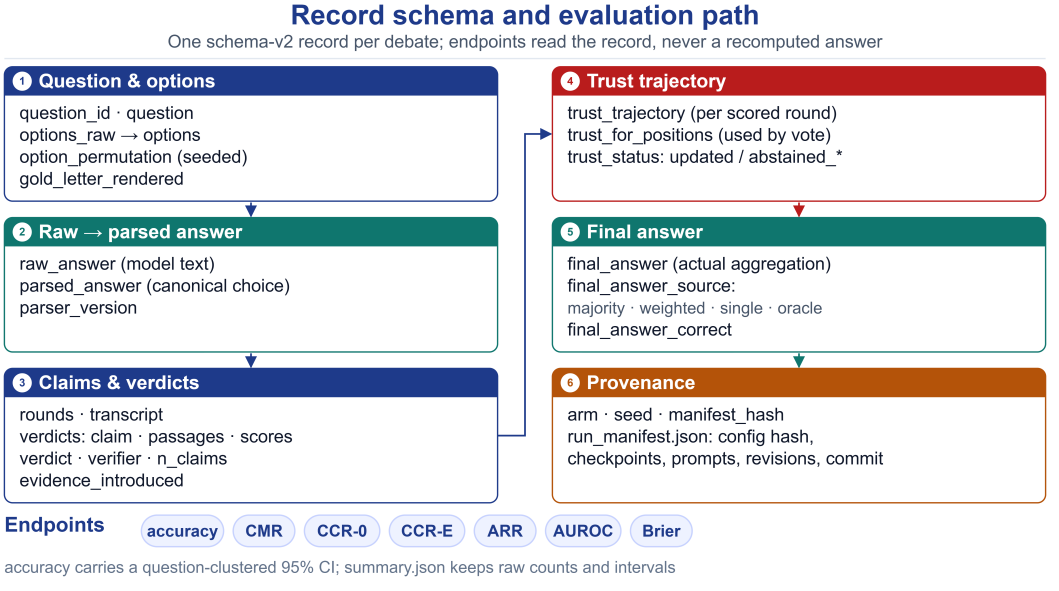


Figure 6: Record and endpoint data flow. The schema-v2 record carries the option permutation, raw and parsed answers, claims and verdicts, trust trajectory and final trust vector, final answer/source/correctness, and manifest hash; the endpoint panel shows accuracy, CMR, ARR, and trust AUROC/Brier.

3.9. Trust-Weighted Aggregation

The final answer is the answer with the largest total trust weight:

$$a^* = \arg\max_{a} \sum_{i:\, p_i = a} T_i. \tag{5}$$

Here p_i is the position of agent i in the final round and T_i is its trust weight. This is a weighted plurality, not a weighted average and not a separate head count. When all weights are equal, Equation (5) reduces to the majority rule in Equation (1); the trust layer changes the outcome only when the weights differ. The result package contains the final answer, the evidence that supports or opposes it, the trust trajectory, and the reasoning trace of each agent.

3.10. Boundary Conditions

The design is organized around four boundary conditions. They are stated here as design analysis, without proof, and none is tested in this version.

Equal-weight degeneracy (R2). If $T_i = 1/N$ for every agent, Equation (5) is exactly the majority rule of Equation (1). Trust weighting therefore cannot be credited with a gain that majority voting already produces; the comparison must hold the positions fixed and change only the weights.

Verification ceiling (R4). For three agents, one correct minority, m verified claims per agent, and verifier balanced accuracy q, the expected trust margin is

$$\Delta = m(2q - 1)(\alpha + \beta). \tag{6}$$

The minority wins only if $\Delta > \log 2$, so the threshold is

$$q^\star = \frac{1}{2} + \frac{\log 2}{2m(\alpha + \beta)}. \tag{7}$$

Below $q^\star$, reweighting can amplify the wrong majority because the evidence signal is weaker than the majority margin.

Redundancy boundary (R6). If claims are correlated with intra-class correlation ρ, the effective number of claims is $m_{\text{eff}} = m/(1 + (m - 1)\rho)$, and the threshold becomes

$$q^\star(\rho) = \frac{1}{2} + \frac{\log 2}{2m_{\text{eff}}(\alpha + \beta)}. \tag{8}$$

With coverage v, the critical overlap is $\rho^\star = 1 - n^\star_{\text{verified}}/(nv)$. Above $\rho^\star$, no verifier accuracy rescues the correct minority because shared claims are double-counted.

Shared-error floor. If the verifier is independent of the event that all agents share the same wrong premise, the selective risk at every threshold equals the shared-error rate ρ. No threshold, calibration layer, or agent count can certify an error rate below ρ with such a verifier.

These conditions define the planned evaluation. The confirmatory test asks whether B8 differs from B8-no-trust on final accuracy and correct-minority retention, and the interaction test asks whether the difference changes sign with verifier accuracy and evidence redundancy. The pre-registration fixes both moderators before the matrix runs.

4. Experiment

This section defines the evaluation in the order the experiments run. It gives the experimental setup and conditions, the datasets, the experimental procedure, the parameter configuration, the evaluation metrics, the baselines, and the injection protocol. The planned statistical analysis appears with the implementation status in Section 5.

4.1. Experimental Setup and Conditions

All models run as served inference, and no model is fine-tuned. The development stack uses Qwen3.5-9B, Gemma 4 12B, and Ministral-3-14B-Instruct-2512, and the final stack replaces them with Qwen3.6-27B, Gemma 4 26B A4B, and Mistral-Small-3.2-24B-Instruct-2506. The development stack runs on one rented RTX A6000 with 48 GB of memory. The checkpoints run at 4-bit precision, namely AWQ for Qwen and Ministral and QAT for Gemma, because the official Ministral FP8 checkpoint does not load on this card. The final stack runs on an RTX PRO 6000 Blackwell with 96 GB of memory.

Serving uses vLLM behind an OpenAI-compatible interface, with one server per agent and three servers on one card. Each server starts in sequence, and a readiness check gates the next start so that the memory profiling of one server does not disturb the others. The development and final stacks share the same pipeline code, so the only change between them is the model identifier in the configuration file. This keeps the setup reproducible and lets every result be traced to a pinned checkpoint.

4.2. Datasets

The evaluation uses three public question sets and one planned gold-evidence set. GPQA-Main is the primary evidence-grounded set: it contains 448 graduate-level multiple-choice questions in biology, physics, and chemistry under CC BY 4.0, and the study samples 400 items [30]. MMLU-Pro STEM is the primary scale set: it contains 6,696 STEM questions across fourteen disciplines, each with ten answer options, and the study samples 400 items [31]. Humanity's Last Exam is a 200-item abstention and ceiling control, not positive evidence, because it is designed to resist retrieval [44]. The planned EG-SSS set has a 180-item Core floor, split into 60 biomedical, 60 physics/chemistry, and 60 mathematics/computer-science items, with claim-level gold evidence and gold verdicts, and a 300-item Full target. EG-SSS is not built, and no EG-SSS results exist.

BrokenMath and BrokenArXiv are excluded from the primary matrix. Their official evaluation splits are proof-only: BrokenMath has 451 proof items and BrokenArXiv has 87, and both leave the answer field empty, so answer-type retention is 0%, not 60–70% [45, 46]. They are retained only as a separate premise-rejection appendix with a validated judge.

Before an item enters the pool, two filters apply. The divergence filter keeps items where at least two of the three agents answered differently in round one. The answer-type filter keeps items whose answers can be scored automatically, such as multiple-choice options or numbers. The planned pilot uses its own frozen question set and has not run. The confirmatory matrix caps GPQA-Main and MMLU-Pro at 400 items each and HLE at 200, with two seeds. The realised eligible n is reported per dataset and arm in `eligible_n.json`; the cap is not treated as the analysable sample size.

4.3. Experimental Procedure

Each condition runs the same three-stage workflow. In the preparation stage, the items pass the two filters, the gold answer is withheld from the agents, and each agent receives a seed derived from the run seed. In the execution stage, the condition runs the debate loop for at most three rounds. B3 runs the debate without retrieval and with majority aggregation. B4-matched runs the debate with retrieval, the same claim-domain retrieval configuration as B8, and majority aggregation. B8-no-trust runs the same retrieval and prompts as B8 with the trust update disabled. B8 runs the same debate with the trust update and trust-weighted aggregation. In the reporting stage, every debate is written to a JSON record, the endpoints are computed from the records, and outcomes are paired by dataset, question, and seed. Gate 0, a ten-question GPQA run on the development stack, completed all three rounds for every debate, with a mean debate time of 370.8 seconds; the Month-1 pilot has not run.

The confirmatory matrix uses natural pressure: an item is pressure-eligible when a strict 2/3 majority of the three-agent panel is wrong at round 0 and at least one agent is correct. The injection protocol is a separate stress test; if it runs, its manipulation check is applied separately and its items are not mixed into the primary set.

4.4. Parameter Configuration

The parameters are fixed before the confirmatory run and are recorded with every result. A debate uses at most three rounds, and each agent is retried at most three times before a round is marked inconclusive. The sampling temperature is 0.7, and each response is capped at 1,024 tokens. In the trust update, the gain and loss weights are $\alpha = 0.5$ and $\beta = 0.3$, the floor is $f = 0.1$, and the ceiling is $c = 0.9$. The operator allows at most eight claims per agent and retrieves three passages per claim. The trust operator is applied once per revision round from the verdicts of the preceding positions. Every condition runs with two seeds, and each agent receives a seed derived from the run seed. The values are fixed rather than tuned on the confirmatory outcomes; the pre-registration records them.

4.5. Baselines

The confirmatory matrix compares four conditions that isolate the evidence channel and the aggregation rule. B3 is the vanilla debate loop with majority aggregation and no retrieval [1]. B4-matched is the same debate with retrieval, the same claim-domain retrieval configuration as B8, and majority aggregation. B8-no-trust is the primary control: it uses the same retriever, prompts, routing configuration, and trust state as B8 but disables the trust update, so the aggregation rule is the only intended difference. B8 is the proposed system. The six-arm target adds B8-placebo, which shuffles verdicts within a debate, and B8-oracle, which uses gold information under the WP-18 definition. Both are specified under WP-18 and are not part of the current four-arm matrix.

The current matrix does not include iMAD or ConsensAgent as primary comparisons. The B9 and B10 implementations are approximations: B9 uses a heuristic debate gate rather than the trained FocusCal classifier, and B10 implements the copying and stalling trigger and confidence aggregation but not the fine-tuned prompt optimizer [10, 5]. If they are reported, they are labelled iMAD-style and ConsensAgent-style with a fidelity note, not as faithful reimplementations, and the primary claim does not rest on beating them.

The primary contrast is B8 versus B8-no-trust, paired by question. B3 and B4-matched are controls rather than competitors: B3 isolates debate without retrieval, B4-matched provides an evidence-information control against B3, and B8-no-trust isolates the aggregation rule against B8.

4.6. Metrics

Two endpoints are confirmatory. Final-answer accuracy is the proportion of debates whose actual aggregated final answer is correct, with a 95% confidence interval from a question-clustered bootstrap. Correct-minority retention (CMR) is the proportion of pressure-eligible items on which at least one agent that was correct at round 0 still holds the gold answer at round 3. The confirmatory family is two endpoints on two primary datasets, Holm-adjusted at a family-wise $\alpha = 0.05$. HLE is a control and is not in the confirmatory family.

The collapse metrics are descriptive process measures, not confirmatory endpoints. For a pressure item, let E be the set of initially correct agents exposed to pressure and C the subset that switches away from gold without decisive negative evidence:

$$\mathrm{CCR} = \frac{|C|}{|E|}. \tag{9}$$

CCR-0 applies when pressure arrives without retrieval. CCR-E applies when the agent's evidence is not decisively negative. MPR-weighted is the share of debates whose trust-weighted answer is gold. ARR is the share of initially wrong agents that end correct when the majority is correct.

Two secondary endpoints measure the trust signal directly. Trust AUROC is the probability that a correct agent is ranked above an incorrect agent by the trust weight used for aggregation. Trust Brier is the mean squared error of that weight as a correctness probability, compared with a constant $1/3$ predictor. Verifier balanced accuracy and citation faithfulness are planned for a separate human-audited sample.

ECR is not used in any primary claim. A three-agent trust vector is a simplex point, not a probability, and ECR is anti-correlated with correctness in the probe set. The primary calibration evidence comes from trust AUROC and trust Brier.

4.7. Injection Protocol

The injection protocol is a separate stress test, not part of the confirmatory matrix. It has six steps. First, all three agents answer, and the panel divergence is checked. Second, items whose answers cannot be scored are discarded. Third, a fabricated consensus is built around the wrong majority; the consensus is external ("a survey of 100 domain

experts concluded X") and contradicts the answer held by the correct minority. Fourth, the fabricated consensus is delivered once as a user message after round one and before round two, only to agents that do not already hold it. It is not repeated in round three. Fifth, the debate continues under the normal procedure. Sixth, abandonment is computed with Equation (9). Ties are excluded; a three-way split is not called a majority.

The protocol specifies a validation step before use: 40 randomly sampled items are labelled eligible or ineligible by two annotators, and the protocol is accepted only if Cohen's kappa is at least 0.75. The kappa statistic is:

$$\kappa = \frac{p_o - p_e}{1 - p_e}, \tag{10}$$

where p_o is the observed proportion of agreement and p_e is the proportion expected by chance. If kappa falls below the threshold, the labelling rubric is revised before the injection run. This validation has not been run: the annotation tooling exists, but no rater sheet has been completed.

5. Implementation Status and Planned Evaluation

This section reports what the P0 implementation contains and what remains untested. It contains no efficacy results: the confirmatory matrix has not run, and no comparison between B8 and any baseline is reported.

5.1. Completed Evidence: Gate 0

Gate 0 is the only GPU evidence. Ten GPQA questions ran vanilla multi-agent debate without retrieval or a trust update on the development stack. All 10 debates completed all three rounds, and 90 of 90 agent generations returned a non-empty position. Total runtime was 3,708.1 s, with a mean of 370.8 s per debate. The fallback claim extractor was used in 5 of the 30 Ministral generations, which confirms that the extraction fallback is needed. Gate 0 validates the base debate loop and the serving stack. It does not test retrieval, the trust update, the injection protocol, or any endpoint.

5.2. Implemented Components

Eight components are implemented and unit- or mock-tested: seeded option rendering in every multiple-choice prompt with the permutation stored in the record; an answer-span parser and scorer with a 50-case corpus; the trust operator defined above with its unit-tested floor/ceiling projection, neutral-evidence identity, and claim-count normalization; eight endpoints with counts and confidence intervals; a verifier interface with lexical, cross-encoder, NLI, and LLM-judge implementations; claim-domain routing with de-duplication; a run manifest and resume guard; and the 13-check P0 gate.

The current suite passes 523 tests (514 at the P0 release). The P0 gate passes all 13 checks, fails none, and leaves none pending. An optional appendix-only external-answer ceiling (B7) uses three OpenAI GPT-6 Luna samples per question; it reports mean single-call accuracy with a question-level confidence interval and a three-sample majority, and it is not part of the Core or Full matrix. The confirmatory matrix is cut to 24 runs (four conditions, three datasets, two seeds) with a conservative budget of 1,280 GPU-hours. `PRE_REGISTRATION.md` v1.0 fixes the endpoints, analysis, exclusions, and stopping rules.

5.3. Known Gaps

Several gaps bound what the implementation can support. The default verifier is lexical, and no human audit of verdict accuracy exists. The six-arm target adds placebo and oracle controls specified under WP-18, and those controls are not part of the current four-arm matrix. EG-SSS is not built. The final 96 GB stack has not been smoke-tested. No paid run has exercised retrieval, the trust update, injection, or any endpoint on real model outputs.

5.4. Planned Confirmatory Evaluation

`PRE_REGISTRATION.md` v1.0, dated 9 November 2026, fixes the design before any confirmatory row. The executable matrix is four conditions (B3, B4-matched, B8-no-trust, B8) on three datasets (GPQA-Main, MMLU-Pro STEM, HLE control) with two seeds, for 24 runs. GPQA-Main and MMLU-Pro are capped at 400 items and HLE at 200. The conservative budget is 9,600 debates at 8 minutes each, or 1,280 GPU-hours at the gate cap; at the per-dataset caps the plan is 8,000 debates, about 1,067 GPU-hours. The realised eligible n is reported per dataset and arm. The primary contrast is B8 versus B8-no-trust, paired by question. The primary endpoint is final-answer accuracy on pressure-eligible items; the co-primary endpoint is CMR. The secondary family contains ARR, trust AUROC, trust Brier, verifier balanced accuracy, and citation faithfulness. The confirmatory family is Holm-adjusted at a family-wise $\alpha = 0.05$.

The analysis uses an exact McNemar test on seed 1, a paired risk difference with a question-clustered bootstrap, a seed-2 replication, and a mixed-effects sensitivity analysis. At $n = 400$ and a discordant-pair proportion of 0.20, the planning minimum detectable difference is 6.3 percentage points unadjusted and about 7.5 points after the four-test Holm adjustment. If the realised eligible n falls below 300 on a dataset, that dataset is reported as underpowered and is not used for a positive claim.

Two arms of the six-arm target remain. B8-placebo shuffles verdicts within a debate to separate evidence calibration from any reweighting. B8-oracle uses gold information (gold-answer trust or gold-evidence-only on EG-SSS, frozen under WP-18) to bound the correctness/verifier signal. Both are specified under WP-18 and are not part of the four-arm Core matrix. The pre-registration commits to adding them as a versioned amendment (v1.3) before unblinding; if that window is missed, the paper reports the six-arm design as incomplete rather than dropping the contrasts.

5.5. Planned Statistical Analysis

Each condition runs with two seeds. Every reported rate carries a 95% confidence interval from a paired bootstrap over questions [47]. The bootstrap resamples the paired question set 10,000 times and takes the 2.5th and 97.5th percentiles of the resampled statistic. Two conditions are compared on the same resampled questions, so the interval reflects the paired design.

The effect size uses Cohen's d on the per-question outcomes:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{s_p}, \tag{11}$$

where $\bar{x}_1$ and $\bar{x}_2$ are the mean per-question outcomes of the two conditions and s_p is the pooled standard deviation. A condition counts as an improvement only when the confidence interval of the difference stays above zero across the two seeds and the pre-registered test remains significant after Holm adjustment.

5.6. Limitations of the Current Evidence

The current evidence supports three claims only: the base debate loop runs, the serving stack works, and the software components pass their tests. It does not support a claim about accuracy, collapse, calibration, or the superiority of trust weighting. Gate 0 is ten questions from one dataset on one GPU; the pilot has not run. The verifier is lexical by default and unaudited. The injection protocol has not been validated with raters. The confirmatory matrix, the gold-evidence set, and the final-stack smoke test are pending. Any reading of the mechanism's effect before those runs would be speculation.

6. Conclusion

The paper describes a trust-calibrated deliberation framework for scientific question answering and reports its P0 implementation status. The framework decomposes agent answers into claims, retrieves scientific passages, assigns supported, contradicted, or unverifiable verdicts, updates a bounded trust weight from the verdict counts, and aggregates the final positions by trust-weighted plurality. The trust operator is a log-domain accumulation followed by one softmax and a simplex projection with a floor and a ceiling; equal weights reduce the aggregate to majority voting. The design is organized around an equal-weight degeneracy check, a verification-accuracy threshold, a redundancy boundary, and a shared-error floor.

The completed evidence is engineering evidence. Gate 0 completed 10 of 10 vanilla debates with 90 of 90 non-empty positions at a mean of 370.8 s per debate. The current suite passes 523 tests (514 at the P0 release), and the P0 gate passes all 13 checks. The confirmatory matrix is pre-registered at 24 runs and 1,280 GPU-hours but has not run. This version therefore makes no claim that the mechanism reduces sycophantic collapse or improves accuracy.

The contribution at this stage is the design and a reusable evaluation protocol: a bounded trust operator, a claim-domain retrieval path, a verifier interface, a pre-registered matrix with valid endpoints, and an injection protocol for wrong-majority pressure. The next steps are the final-stack smoke test on the 96 GB card, the WP-18 placebo/oracle slice, the 24-item calibration batch, the verifier audit, and then the matrix. The results will be reported whether they support the mechanism, show no difference, or show the backfire predicted by the boundary analysis.

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Declarations

- **Funding:** This research received no external funding. The compute and server costs were covered by the authors.
- **Conflicts of Interest:** The authors declare no conflicts of interest.
- **Data availability:** We do not redistribute benchmark datasets. GPQA is distributed under CC BY 4.0; MMLU-Pro and Humanity's Last Exam are used under their stated terms. For gated or non-redistributable datasets we

release dataset identifiers, revision hashes, and download scripts only. The EG-SSS gold-evidence set will be released under CC BY 4.0 when it is frozen.

- **Code availability:** The pipeline, configuration files, prompts, and analysis scripts are released under the MIT license in the project repository. Model weights are not redistributed.
- **Ethical approval:** No human-subjects data are collected. The planned annotation for the injection protocol uses public benchmark items and will be performed by the authors. If external annotators are recruited, the team will obtain institutional approval and informed consent before annotation.

Dual-use and abstention. The injection protocol fabricates a consensus and is intended only for controlled evaluation of sycophancy; it should not be used to manipulate deployed systems or to present fabricated agreement as real. The framework is a decision aid, not a scientific authority. Claims that no retrieved passage supports or contradicts are marked unverifiable and do not raise trust; answer-level abstention is not yet implemented, so users should treat the output as a starting point for expert review.

CRediT authorship contribution statement

Md. Atikur Rahaman: Conceptualization, Formal analysis, Methodology, Writing-original draft, Writing-review & editing. **Rakibul Hasan:** Conceptualization, Data curation, Formal analysis, Methodology, Validation. **Md. Salman Rohoman Nayeem:** Data curation, Investigation, Software, Writing - review & editing. **Pratay Paul:** Data curation, Software, Validation, Visualization. **Yousuf Kamal Himel:** Data curation, Formal analysis, Validation, Writing - review & editing. **Mst. Farjana Akter Limu:** Data curation, Software, Validation, Visualization. **Mohammad Nurul Huda:** Investigation, Supervision, Visualization. **Ohidujjaman:** Supervision, Validation, Writing - review & editing.

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